

# Vyomini Joshi

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</> LeetCode | GitHub

## PROFILE

Pre-final year Computer Science student with a strong foundation in data structures, algorithms, and core software design, demonstrated by solving 350+ problems on LeetCode with a peak contest rating of 1526. Gained hands-on engineering experience spanning systems-adjacent networking (libpcap/C++), Machine Learning applications, and Full-Stack development. Practical and collaborative, with a proven ability to build optimized, end-to-end software architectures and eager to contribute to software engineering teams.

## EDUCATION

### National Institute of Technology, Surat

2024 – 2028

B.Tech – Computer Science and Engineering (CGPA: 8.42 Till 4th Semester)

Surat, India

- **Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence, Information Security
- **Academic Highlights:** CBSE Class XII: 94.4% | CBSE Class X: 95.6%

## WORK EXPERIENCE

### Development Contributor (Under Senior Mentorship) | DeepWire DPI

In Progress

- **Tech:** C++, Go, CMake, libpcap, BPF, Concurrency, IPC
- Engineering the **ingress handler layer** of a high-performance Deep Packet Inspection (DPI) system engineered for real-time network traffic analysis.
- Implementing raw packet capture mechanisms using libpcap in promiscuous mode, targeting memory footprint reduction and runtime optimization at the entry layer.
- Applying kernel-level Berkeley Packet Filter (BPF) rules to isolate HTTPS traffic, mitigating critical pipeline processing bottlenecks at the ingress stage.
- **GitHub:** [DeepWire-DPI-Smart-Network-Traffic-Analyser](#)

### Google Winter of Code 2026 | Rabuste Café

Dec 2025 – Feb 2026

- **Tech:** React.js, Node.js, Express.js, Tailwind CSS, Firebase, LLM APIs
- Developed a full-stack, production-ready e-commerce platform for a local commercial application as part of GWoC, contributing **primarily to frontend architecture** alongside focused backend engineering.
- Architected modular, responsive, and reusable user interface layouts in React.js, applying clean component hierarchy rules to eliminate performance layout shifts.
- Identified and resolved runtime state race conditions and UI rendering latency gaps through intensive environment debugging.
- **Live Deployment:** [gshock12.vercel.app](#) | **GitHub:** [GShock](#)

## PROJECTS

### CodeBoard | MongoDB, Express.js, React, Node.js, Socket.IO, JWT

- Developed CodeBoard, a MERN-based collaborative coding platform featuring real-time multi-user code synchronization.
- Enabled live collaboration using Socket.IO to deliver instant, low-latency code updates across connected clients.
- Implemented a secure authentication workflow utilizing JWT for robust user login and access control management.
- Optimized application state management and client-server communication channels for seamless, real-time user interactions.

- **GitHub:** [CodeBoard](#)

### **Pocket Pilot** | *Python, Scikit-learn, Pandas, NumPy, Data Mining*

- Engineered an end-to-end ML application for multi-stage financial data mining, expense forecasting, and recommendation generation.
- Processed transaction-level datasets utilizing advanced feature engineering to uncover latent user behavior patterns.
- Implemented K-Means clustering algorithms to segment spending behavior models into reliable predictive matrices.
- Optimized model performance strategies to facilitate accurate early detection of user financial anomalies before month-end.
- **GitHub:** [PocketPilot](#)

### **CardioVision** | *HTML, CSS, JavaScript, MediaPipe, Signal Processing*

- Built a complex, browser-based system executing contactless heart rate extraction via remote photoplethysmography (rPPG).
- Processed multi-dimensional landmark coordinates dynamically to isolate regions of interest for continuous wave isolation.
- Developed numerical pipelines (CHROM, detrending, DFT) to capture and analyze real-time mathematical pulse metrics.
- Designed an interactive medical dashboard featuring automated health metrics analysis and automated JSON export tooling.
- **Live Deployment:** [rppg-js.vercel.app](#) | **GitHub:** [rppg\\_js](#)

## **SKILLS**

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**Languages :** C++, C, Python, JavaScript, SQL (MySQL)

**Systems & Networking:** Computer Networks, TCP/IP, Linux Basics, libpcap, Berkeley Packet Filter (BPF), Concurrency, IPC

**Machine Learning & Mining :** Scikit-learn, Feature Engineering, Regression, Classification, K-Means Clustering, Data Mining, Model Evaluation

**Software Engineering & Tools:** Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Git, GitHub, VS Code

**Web Development :** React.js, Node.js, Express.js, MongoDB, Tailwind CSS, HTML, CSS, Firebase, Socket.IO

**Data Systems :** NumPy, Pandas, Matplotlib, Data Analysis, Visualization Tools, Statistics, Probability

## **ACHIEVEMENTS**

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- **Web Wonders 2025** – Finalist
- **Google Winter of Code 2026** – Winner (Developed production-ready, client-facing web implementation).
- **INSPIRE-MANAK Scholar** (Government of India)

## **POSITIONS OF RESPONSIBILITY**

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- **Representative** – Nexus, NIT Surat (Aug 2025 – Present)
- **Executive** – E-Cell, NIT Surat (Sep 2025 – Feb 2026)
- **Volunteer** – NSS (May 2025 – Jan 2026)